BA820 – Project M4

Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Parisha Patel

1. Refined Problem Statement & Focus (~0.5 page)

The domain question that I explore in this milestone is: *Do directors characterized by distinct casting age-gap profiles also exhibit structured collaboration patterns, indicating persistent casting ecosystems rather than isolated creative decisions?*

This central question is the same as M2, but now there is no longer a need to find clusters, but to demonstrate their strength and permanence over time. At previous milestones, the observed clusters of directors were meaningful in terms of their structural casting behavior. I specifically test the assumption in M4 by methodologically refining. Two assumptions are checked. First, the algorithmic randomness of clustering does not occur due to randomness. My testing using several random seeds and hierarchy checking with hierarchical clustering helps to verify that the segmentation of the ecosystem is stable and structurally consistent. Second, the ecosystems of the directors are temporal. In part, I disprove this assumption by classifying careers into early and late, and this is that clusters are internally stable, but there are also directorswhot change their categories over phases. M4, therefore, amplifies the original framing as it shifts towards structural validation based on robustness as opposed to descriptive segmentation.

Unlike M3, which emphasized cross-level integration across regimes, packaging, and director ecosystems, M4 isolates the director-ecosystem mechanism and tests its internal structural validity. The focus shifts from demonstrating linkage across levels to evaluating whether director ecosystem segmentation remains stable under alternative specifications, including multiple random seeds, hierarchical clustering cross-checks, and temporal segmentation. This reflects a conceptual progression from breadth of integration (M3) to depth of structural validation (M4).

2. EDA & Preprocessing: Updates (~0.5 page)

Previous EDA indicated that director-level age-gap behavior had high heterogeneity, and the distribution of romantic couples per director was very skewed. I used the M2 preprocessing decision to eliminate directors whose number of recorded couples is too small because it is easy to over-interpret sparse cases, and, therefore, I used ecosystem profiles to represent repeated casting behavior and not a single-film noise. Director-level characteristics, median age gap, dispersion, proportion of large gaps, and normalized collaboration concentration were normalized before using clustering to avoid domination of scale, in line with PCA and KMeans assumptions of distance measures.

In M4, there were no new variables added, but preprocessing was also expanded to allow refinement of methods. First, I standardized the parameters of global configurations (e.g., k, seed values) to be able to reproduce the same result on repeated runs. Second, I used temporal segmentation that divided the filmography of each director by the median release year and manually used segment-level filtering thresholds to avoid low within-phase sample sizes. Such updates were required to measure robustness and temporal persistence without making any changes to the fundamental feature structure.

3. Analysis & Experiments (~2 pages)

This advancement enhances my director-ecosystem analysis provided in the previous section by rendering results of the clustering more defensible due to systematic tuning, robustness verification, and cross-method comparison. The business/domain objective will be the same: are directors displaying structured types of eco-systems, combinations of age-gap casting habits and collaboration focusing, not idiosyncratic, film-by-film change. In this case, unsupervised clustering is the right choice, as there have been no ground-truth labels of the types of ecosystem, and it needs segmentation to be interpreted instead of being predicted.

3.1 Feature representation and why clustering is appropriate

I build director-level characteristics that are a summary of two processes: (1) age-gap behavior (central tendency and dispersion) and (2) collaboration ecosystem structure with normalized recurrence proxies, which minimize confounding due to director volume (i.e., prolific directors are inherently more likely to repeat). Very sparsely observed directors may give unsteady profiles, and to guarantee that the clustering is not performed by isolated outliers, I use a minimum-couples filter. This is due to the fact that features are not on the same scale, thus I would standardize them first and subsequently cluster them after that, such that no one measure predominates the distance calculations. Such an arrangement corresponds to the analysis objective: in case ecosystems exist, directors ought to create groups in a standardized feature space.

3.2 Selecting the number of clusters (k): diagnostics and expectations

K-Means requires choosing k, so I treat this as a model-selection problem rather than a single arbitrary choice. I ran two standard diagnostics taught in class:

- **Elbow/inertia curve** to evaluate diminishing returns in the within-cluster sum of squares (see **Figure A1**).
- **Silhouette sweep** to assess cohesion vs separation across candidate k values (see **Figure A2**).

Silhouette reached its maximum value at k = 2, whereas k = 3 offered poorer separation, but possibly more meaningful interpretation (three-way ecosystem typology instead of a dichotomy of the two). I had expected that there would be a maximum of a few clusters (2-4) that would be the most realistic due to the scale and noise of cultural-production data.

Decision: I use k = 3 as the central specification since it is interpretable (as it has multiple ecosystem types) but can be repeated (as proven below). I use silhouette as a diagnosis- not the only criterion of selection- because the optimal k can be determined by tradeoffs between separation and interpretive usefulness.

3.3 Stability across random seeds (algorithmic robustness)

A key refinement in M4 is demonstrating that ecosystem assignments are not artifacts of one lucky initialization. KMeans can vary across runs due to random centroid starts, so I reran

KMeans at k = 3 across many random seeds and quantified agreement using Adjusted Rand Index (ARI) (see Table A1). This experiment directly tests: if ecosystems are a real structure, assignments should be consistent across seeds.

What worked: The overall stability was high (mean ARI ≈ **0.84**), which implies that the clustering solution is not sensitive to algorithmic noise but quite robust to random initializations.
What surprised me: Not all of the runs were identical (ARI minimum of about ~0.49) because it reminded me that to be stable on average, the clusters must be moderately separated, and not identical all the time. This was a motivation for cross-method checks as opposed to a single measure.

3.4 PCA as an interpretation aid (not a clustering method)

In order to make the cluster structure visually coherent, I projected the standardized space of the features into two dimensions with the help of PCA to allow interpretation (see Figure A3). The PCA is relevant in this case since it is useful in diagnosing the presence of clusters in the data, which depict broad structural directions (e.g., extremity + recurrence) and not random artifacts. I underline the fact that PCA is employed just to visualize, but not to provide allocation.

3.5 Cross-method validation: hierarchical clustering (Ward)

In order to match the course's focus on comparison across clustering methods, I hierarchically clustered the same standardized feature space with Ward linkage and sliced the dendrogram at k = 3, followed by comparison of assignments to KMeans using ARI and a cross-tab (see Table A2 and Figure A4).

Why this helps: In case two distinct clustering paradigms (partitioning vs agglomerative) meet, this supports the argument that ecosystem structure is not a method effect.
What worked: Ward and K-Means had an agreement that was significantly greater than chance, which is evidence of structural consistency.
Challenge: HCs are sensitive to early merges and fail to re-optimize again as KMeans does, so it should not be expected that they reach an imperfect agreement; convergence on coarse structure, and not the same labels, when this occurs.

3.6 Temporal segmentation: testing persistence across career phases

I also tested the permanence of ecosystem identity through time by dividing the filmography of each director into early vs late career (median release year) and repeating clustering in each segment (see Table A3 and Figure A5). This contends with the conceptual weakness of M2: full-career aggregation implicitly presupposes that directors are behaviorally fixed.

What I learned: Clusters formed in the beginning were extremely stable, and late clusters were a little less stable. A significant proportion of directors switched membership in the clusters between phases. This implies that ecosystems are not necessarily fixed, but structured; some directors can evolve, and some movement can be an indication of smaller sample sizes of segments.

Overall takeaway: The ecosystem segmentation is no single descriptive finding; tuning diagnostics, seed resilience, PCA-based interpretability tests, cross-method validation, and cross-temporal stress tests all support it. Notably, strength does not mean being causal. Ecosystem clusters are constant over seeds and techniques, but they are descriptive structures yielded by observed patterns of casting. The clustering algorithms determine a common pattern of age-gap and collaboration behavior, but do not determine whether collaboration concentration is the source of age-gap magnitude or the other way round. Differentiating between structural stability and causal interpretation is a way of making sure that the conclusions are within the scope of unsupervised learning.

4. Findings & Interpretations (~1 page)

The sophisticated analysis reinforces the finding that director casting behavior sexual activity is an organized behavior as opposed to haphazard differences in motion pictures. Directors continuously cluster into separate ecosystems that are defined by age-gap tendencies and collaboration concentration. The observation that these clusters are very stable during repeated model runs indicates a meaningful structural differentiation of the segmentation and not algorithmic noise.

Among the major findings is the fact that age-gap conduct and the collaboration framework seem to be closely connected. Directors with a higher proportion of age gap in their movies also have a more focused collaboration structure, whereas those with more moderate age gaps have a larger actor base. This trend means that the age differences in love relations are not just solitary artistic decisions but can be supported by repetitive working relations and internalized production standards. In terms of industry, this implies that the results of representation can be caused by the existence of stable casting ecosystems, not by the decisions about the project that are made on an individual basis. Time is troped with temporal segmentation. Although the structure of the ecosystem is consistent throughout the career stages, a significant proportion of directors change cluster memberships between the early and late periods. This means that casting ecosystems are not fixed and are structured. This seems to be the behavior of some directors, as their casting priorities seem to be changing with time, possibly as a response to shifting audience expectations, industry demands, or personal artistic growth. Notably, movement is partially related to smaller segment-level samples, which implies that not every reclassification depicts a substantive behavior change.

Taken together, these results indicate that the issue of age-gap inequality in cinema has its roots in repetitive systems of collaboration, which are resistant to the modelling of variation and can, however, evolve. In the case of the studios, creative executives, and diversity programs, this means that the issue of representation might have to be addressed at the network level, rather than by a few movies.

Appendix

Shared GitHub Repository (Required)

- https://github.com/patelparisha/ba820-b1-team10
- Colab Link
- My files exist in the M4 subfolder of the Parisha folder in the Milestones branch of the git repository.

Supplemental Material (Highly Recommended)

Figure A1. Elbow curve showing within-cluster sum of squares (inertia) across candidate k values. The curve shows diminishing marginal reduction beyond k = 3.

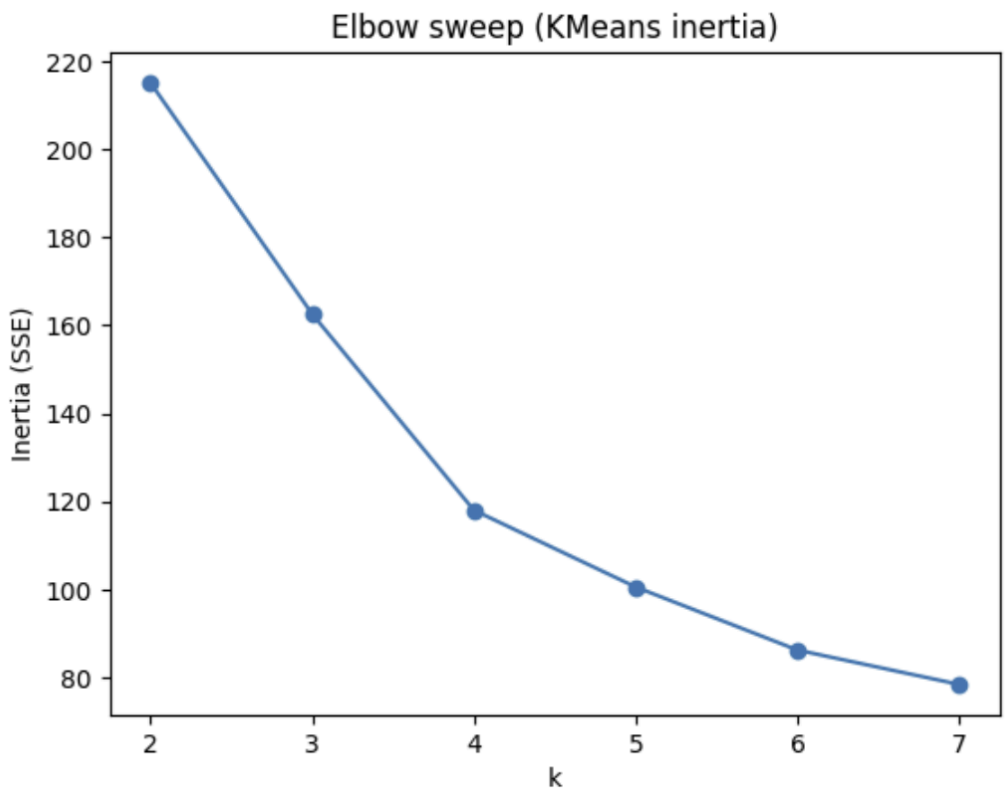

Figure A2. Silhouette scores across k = 2–7. Separation peaks at k = 2, while k = 3 offers slightly lower separation but improved interpretability.

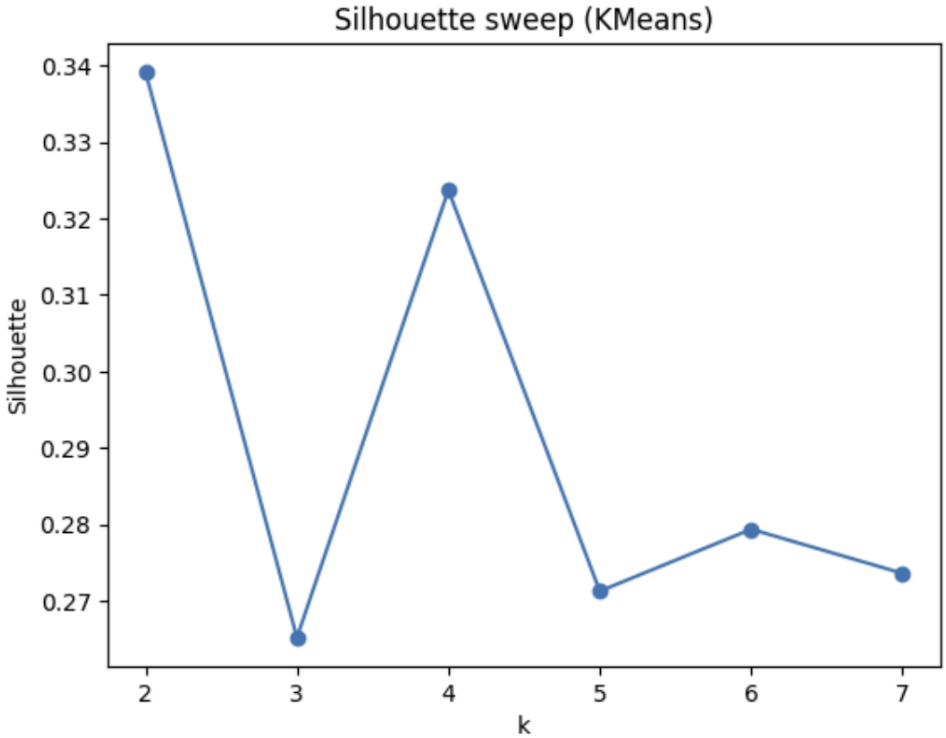

Table A1 – Seed Stability Summary - Stability of KMeans assignments across 60 random initializations (Adjusted Rand Index).

```
{'k': 3,
 'n_runs': 60,
 'ARI_mean': 0.8444574137407226,
 'ARI_std': 0.20443115865039083,
 'ARI_min': 0.4875304336382181,
 'ARI_max': 1.0}
```

Figure A3. K-Means clusters projected into a two-dimensional PCA space. PCA is used for visualization only; cluster assignments are derived from the standardized feature space.

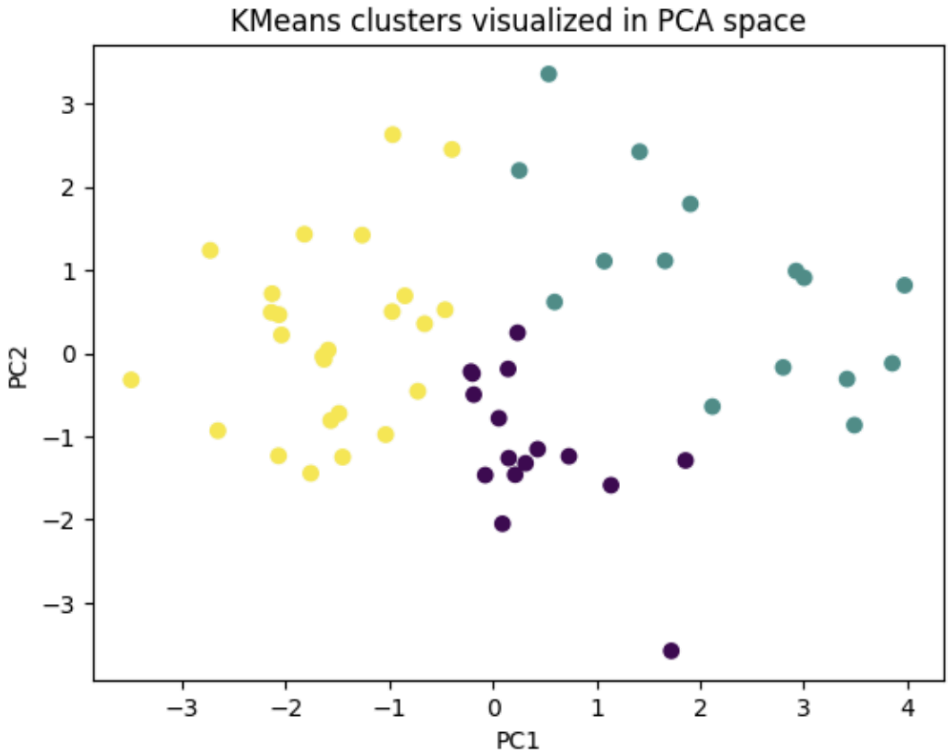

Table A2 – Hierarchical vs KMeans Cross-Tab - Cross-method comparison between hierarchical (Ward) and KMeans clustering assignments (row-normalized). An agreement above chance supports structural consistency.

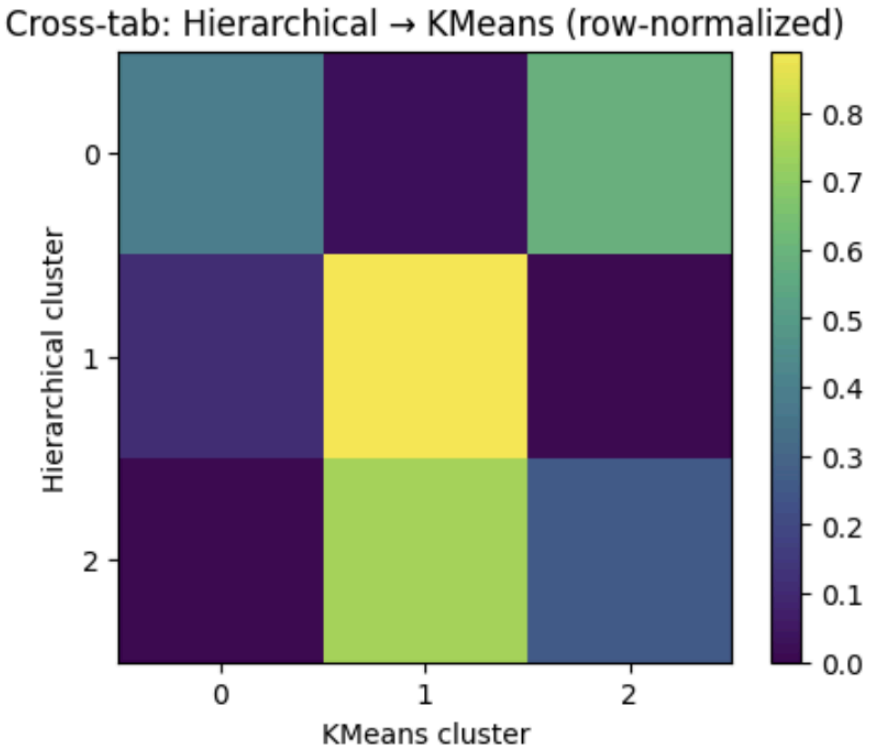

Figure A4. Ward hierarchical clustering dendrogram (truncated view) illustrating hierarchical merge structure.

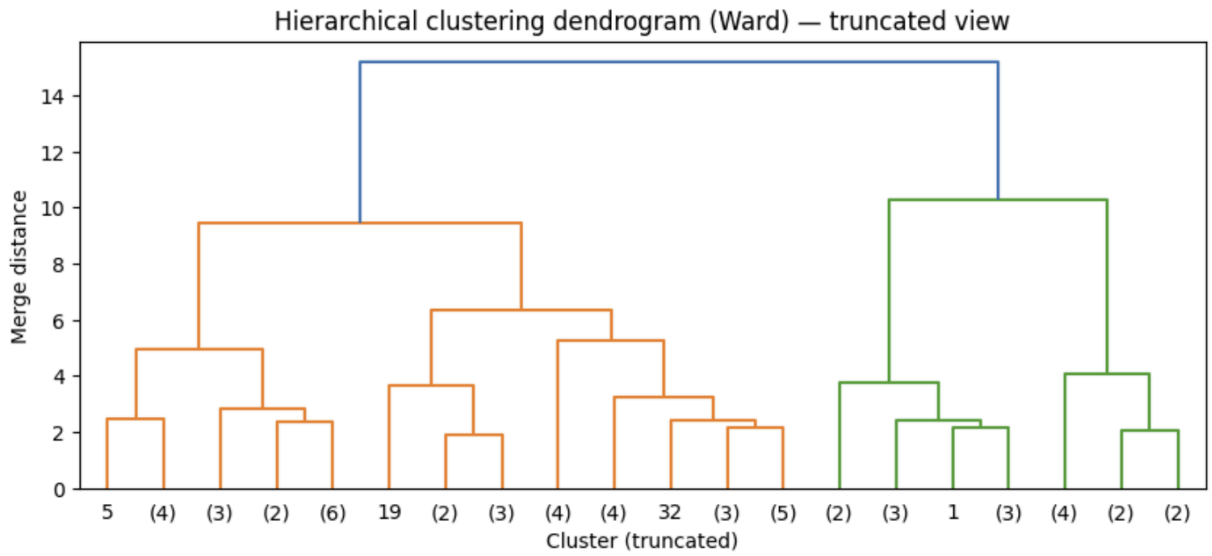

Table A3 – Temporal Movement Matrix - Transition matrix showing movement of directors between early and late career clusters.

```
Directors with both segments: 26
Movement rate (early->late): 0.5769230769230769
```

label_late	0	1	2
label_early			
0	0.500000	0.166667	0.333333
1	0.111111	0.555556	0.333333
2	0.454545	0.272727	0.272727

Table A4 – Segment Stability Comparison - Segment-level clustering stability (ARI) for early and late career subsets.

```
({'k': 3,
  'n_runs': 50,
  'ARI_mean': 0.8879924814302997,
  'ARI_std': 0.19305088205386156,
  'ARI_min': 0.4411454465274279,
  'ARI_max': 1.0},
 {'k': 3,
  'n_runs': 50,
  'ARI_mean': 0.7375513099082723,
  'ARI_std': 0.28856764657397815,
  'ARI_min': 0.29955430938879707,
  'ARI_max': 1.0})
```

Table A5 – Final Summary Table - Consolidated model comparison summary across diagnostics and robustness checks.

	k_selected	silhouette_best_k2	stability_full_ARI_mean	stability_full_ARI_std	movement_rate	stability_early_ARI_mean	stability_late_ARI_mean	n_overlap_directors
0	3	0.339188	0.844457	0.204431	0.576923	0.887992	0.737551	26

Process Overview

The analytical pipeline begins with loading the Hollywood Age Gap dataset directly from a public GitHub source to ensure transparency and reproducibility. After importing the data, I convert key variables such as release year and age difference into numeric format and remove observations missing essential identifiers. This ensures that director-level feature construction is based on valid and interpretable records rather than incomplete entries.

Next, I aggregate film-level observations into director-level behavioral profiles. These features capture two primary dimensions: age-gap structure (median, mean, dispersion, and proportion of large age gaps) and collaboration structure (normalized repeat actor concentration and actor diversity). To avoid unstable clustering driven by sparse cases, I filter out directors with fewer than a minimum number of observed couples.

Because clustering relies on Euclidean distance, I standardize all selected features to ensure that variables measured on different scales contribute comparably. I then apply K-Means clustering to identify latent director "ecosystem" types. Model selection is guided by silhouette scores and elbow diagnostics, balancing statistical separation with substantive interpretability. After selecting k = 3, I conduct robustness checks by testing seed stability usingthe Adjusted Rand Index, cross-validating results with hierarchical clustering (Ward linkage), and verifying consistency under PCA-based dimensionality reduction.

Finally, I split directors into early and late career phases using a director-specific median release year and repeated the clustering procedure within each segment. Comparing segment-level stability and movement rates allows me to evaluate whether ecosystems are persistent or phase-dependent. Throughout the process, I interpret clusters as descriptive structural patterns rather than causal relationships.

Use of Generative AI Tools

1. https://chatgpt.com/share/699ce682-84d0-8013-b8dd-d6ba903afc94